



Marketplace Performance & Customer Experience Analysis

Data Analytics Hackathon

From Data to Better Customer Experiences

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Data Analytics Hackathon Submission



Why This Analysis Matters?

Olist operates a large and growing e-commerce marketplace in Brazil, connecting customers, sellers, products, payments, and logistics partners.



Delivering a consistent and high-quality customer experience is challenging.



Performance varies across delivery, reviews, sellers, product categories, payments, and regions.



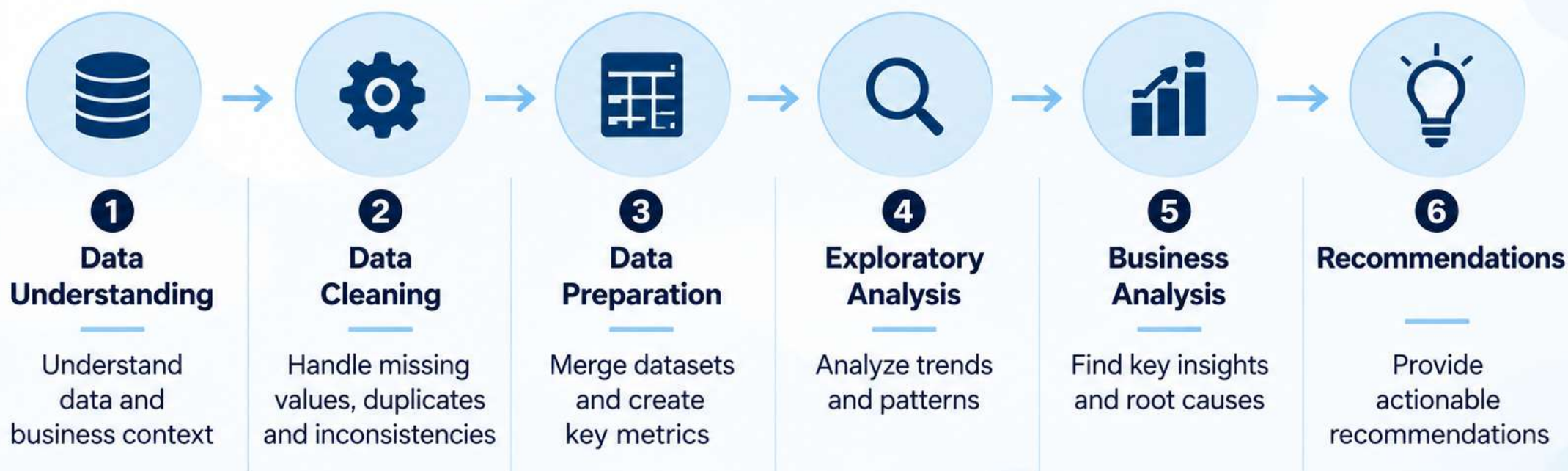
Olist needs to understand which factors are most strongly associated with customer satisfaction and dissatisfaction.



The goal is to move beyond isolated metrics and develop a consolidated, evidence-based understanding of marketplace performance.

Our 6-Phase Analytical Approach

A structured process to turn raw data into actionable insights



From Data to Decisions → Better Customer Experience → Stronger Marketplace

Olist E-commerce Dataset

~ 100,000 orders (Sep 2016 – Oct 2018)



9

Datasets



100K

Orders



2+ Years

of Data

Key Entities



Customers



Orders



Order Items



Sellers



Payments



Reviews



Geolocation



Category
Translation

Important Variables

- Order status
- Purchase date
- Delivery date
- Review score
- Product category
- Payment type
- Customer state
- Seller location
- Product price
- Freight value

Real data. Real customers. Real business impact.

From Raw Data to Analysis-Ready Data

Ensuring data quality, consistency, and reliability

-  Handled missing values
-  Removed duplicates and invalid records
-  Standardized data formats
-  Converted date columns to datetime
-  Merged relevant datasets
-  Created key business metrics
(e.g., delivery days, delivery status, revenue)



Clean Data



Reliable Analysis



Better Decisions

Strong Growth in Orders and Revenue

Monthly Order Trend



Monthly Revenue Trend

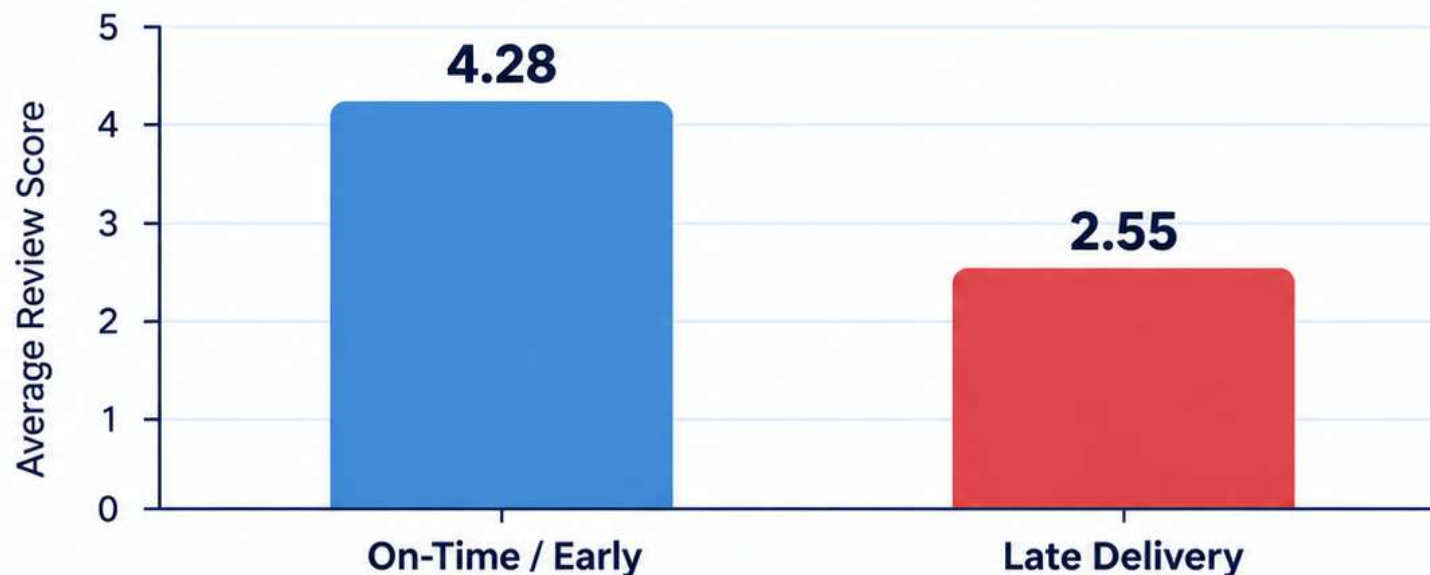


Key Finding

Olist experienced strong growth in both order volume and revenue throughout 2017, reaching its highest levels in late 2017 and early 2018. After this rapid growth phase, performance remained relatively stable at a high level during 2018.

Delivery Delays Strongly Impact Customer Satisfaction

Average Review Score by Delivery Performance



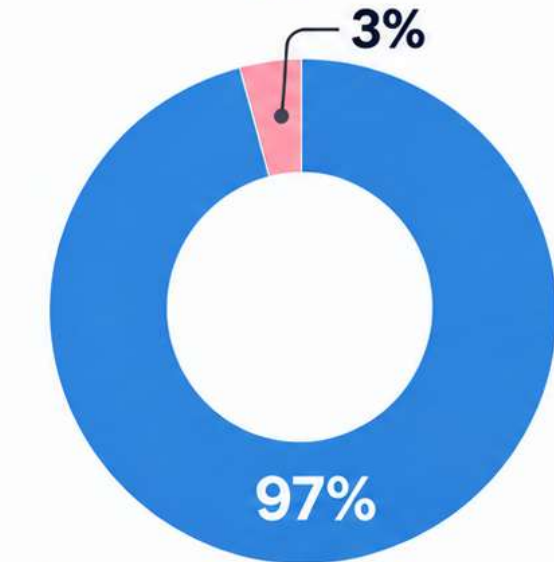
1.73
Point Drop
in Review Score



Delivery delays are the strongest observed factor associated with lower customer satisfaction.

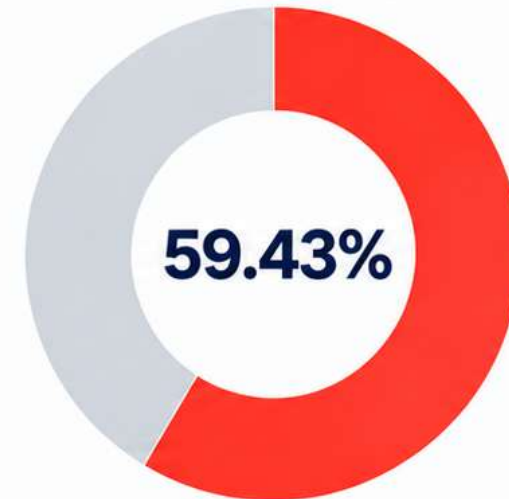
Customer Retention Opportunity

Customer Type Distribution



- One-Time Customers (97%)
- Repeat Customers (3%)

Revenue Concentration by Customer Segment



- Top 25% Customers (59.43% of revenue)
- Other 75% Customers (40.57% of revenue)



Most customers are one-time buyers, while a relatively small group of high-value customers generates a significant share of revenue.

What Drives Low Customer Satisfaction?

Multiple factors contribute, but delivery delays are the strongest observed factor associated with low customer satisfaction.



Other Important Insights



Geographic Performance

- Delivery performance varies across states
- Some regions have higher delay rates and lower satisfaction



Seller Performance

- Performance varies significantly across sellers
- A small group of high-performing sellers contribute a large share of revenue



Product Category Differences

- Categories show variations in sales, pricing, and customer experience
- Some categories require operational attention



Payment Behavior

- Credit cards account for the majority of revenue
- Installments support higher-value purchases
- Weaker evidence of direct impact on satisfaction

Recommendations

Based on these findings, we recommend five priority actions.

1



Implement Delivery Monitoring & Early-Warning Systems

Track delivery performance in real time and set up early-warning systems to identify and prevent potential delays.

2



Improve Logistics in High-Delay Regions

Strengthen logistics operations and focus on high-delay regions to reduce delivery times and improve reliability.

3



Monitor Seller Dispatch & Fulfillment Performance

Track seller dispatch times and fulfillment performance more closely, and support underperforming sellers to improve service standards.

4



Increase Repeat Purchases Through Campaigns & Loyalty Strategies

Use targeted second-purchase campaigns and loyalty strategies to encourage customers to buy again.

5



Protect High-Value Customers

Offer personalized offers and improve service recovery to retain high-value customers and enhance their overall experience.



These actions will help Olist improve customer satisfaction, build loyalty, and drive sustainable growth.

Key Recommendations for Olist



Practical actions to improve customer experience and drive sustainable growth.

- 1 | Implement delivery delay monitoring and early-warning systems.
- 2 | Improve logistics performance in high-delay geographic regions.
- 3 | Monitor seller dispatch and fulfillment performance more closely.
- 4 | Increase repeat purchases through second-purchase campaigns and loyalty strategies.
- 5 | Protect high-value customers with personalized offers and improved service recovery.

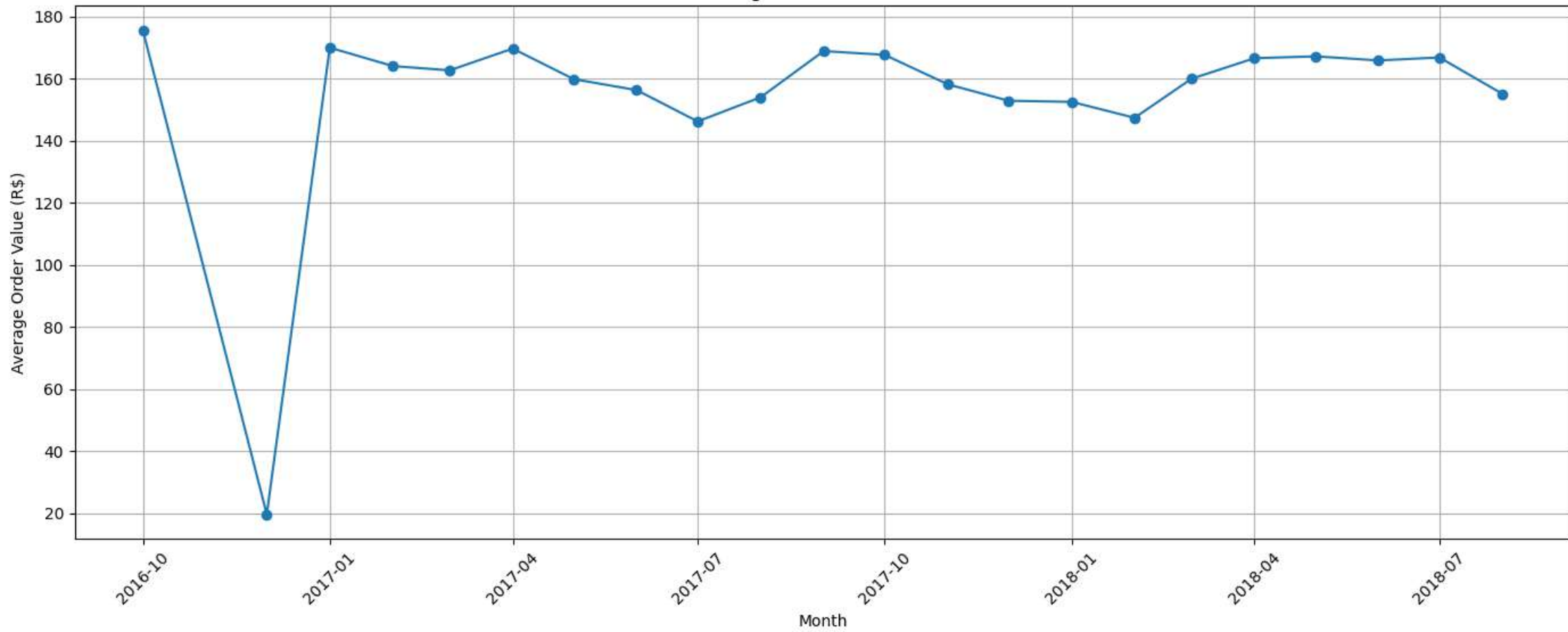
Building a Stronger Olist Marketplace



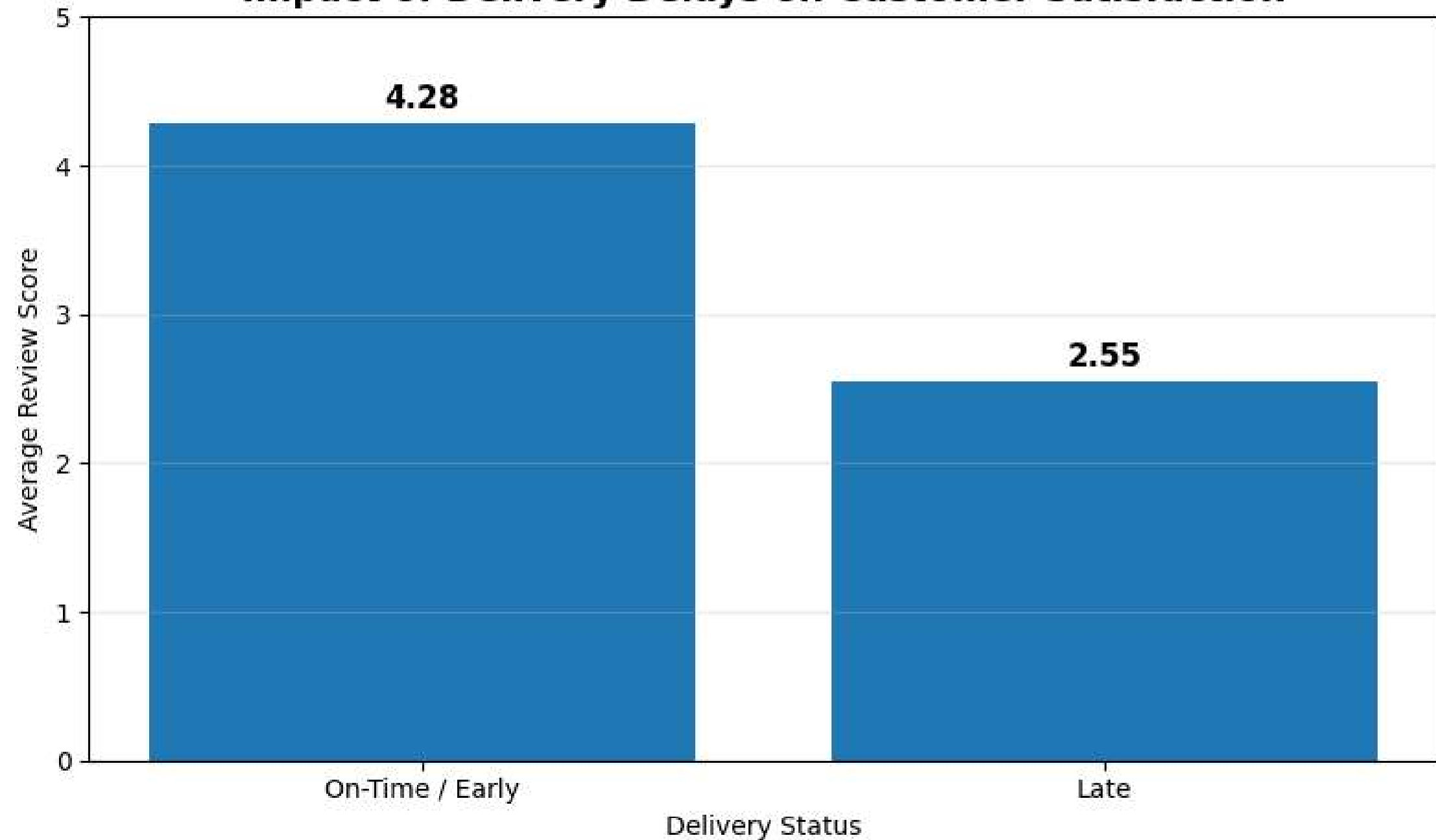
Improving delivery reliability and customer satisfaction
is key to long-term success.

Thank You!

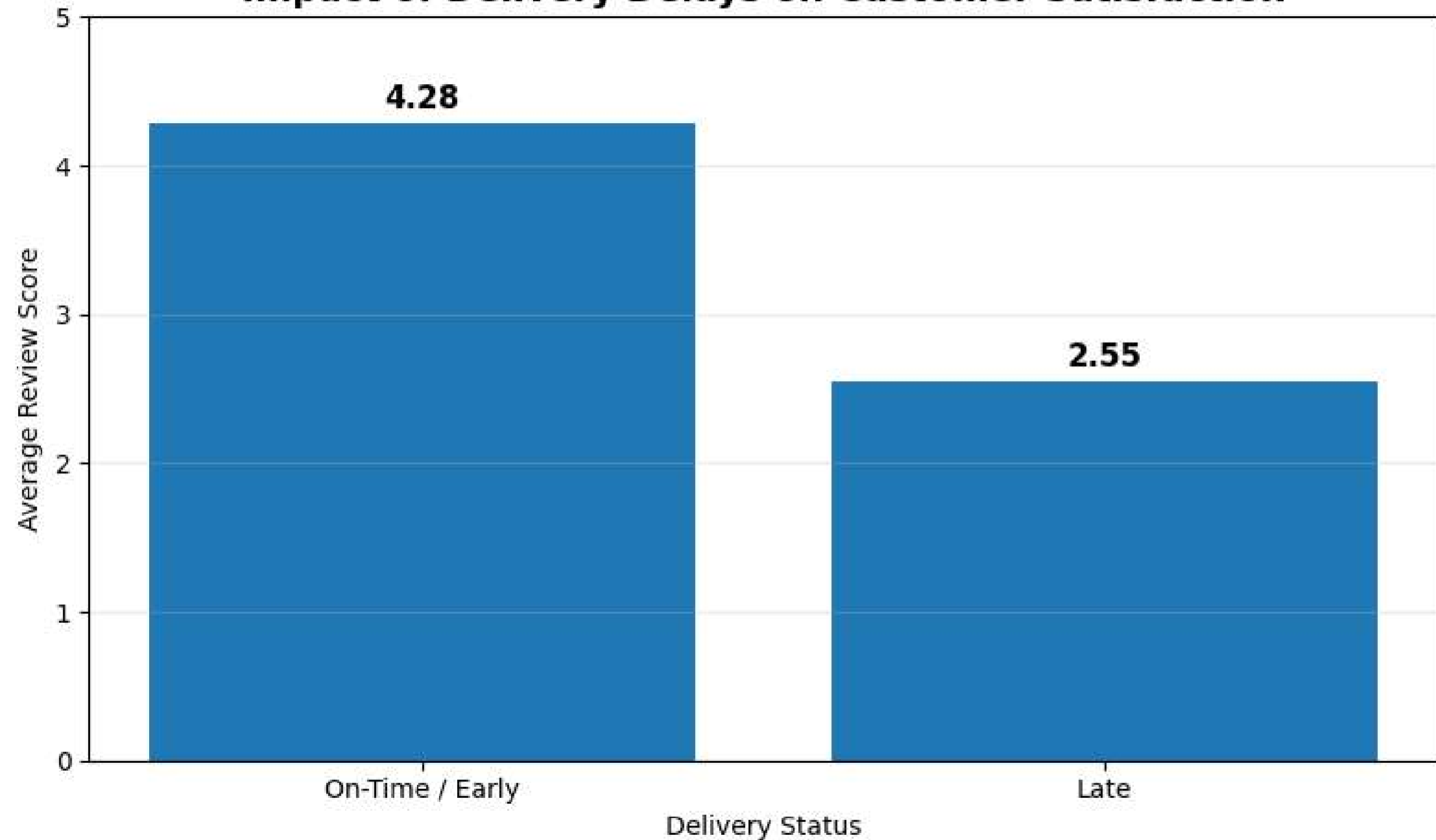
Average Order Value Trend

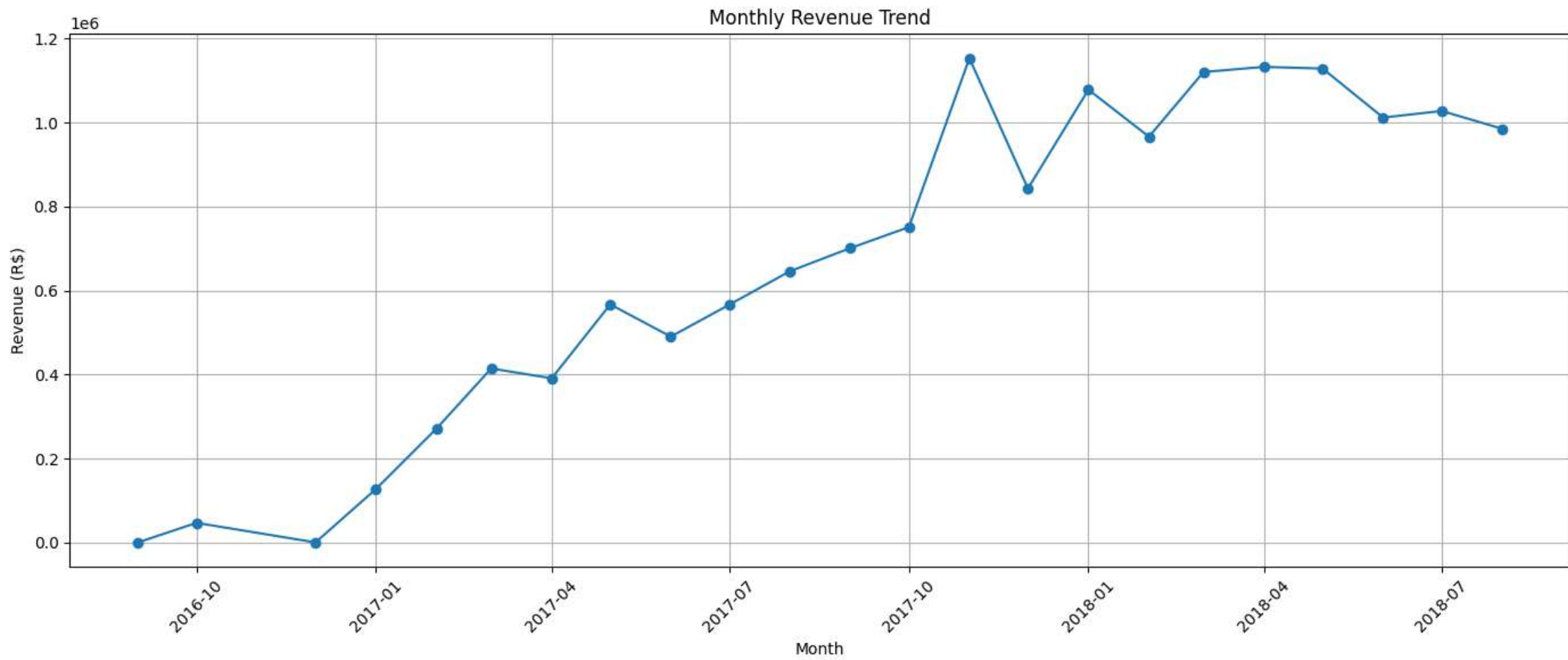


Impact of Delivery Delays on Customer Satisfaction

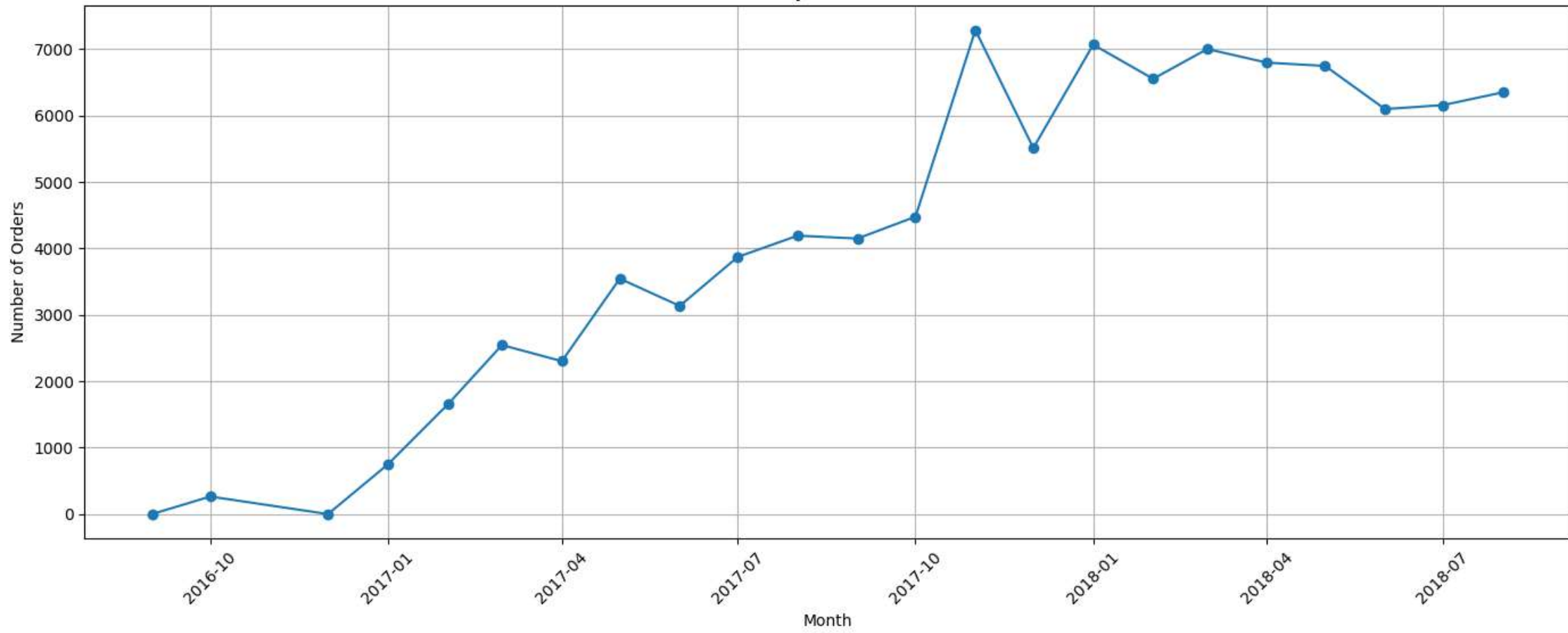


Impact of Delivery Delays on Customer Satisfaction

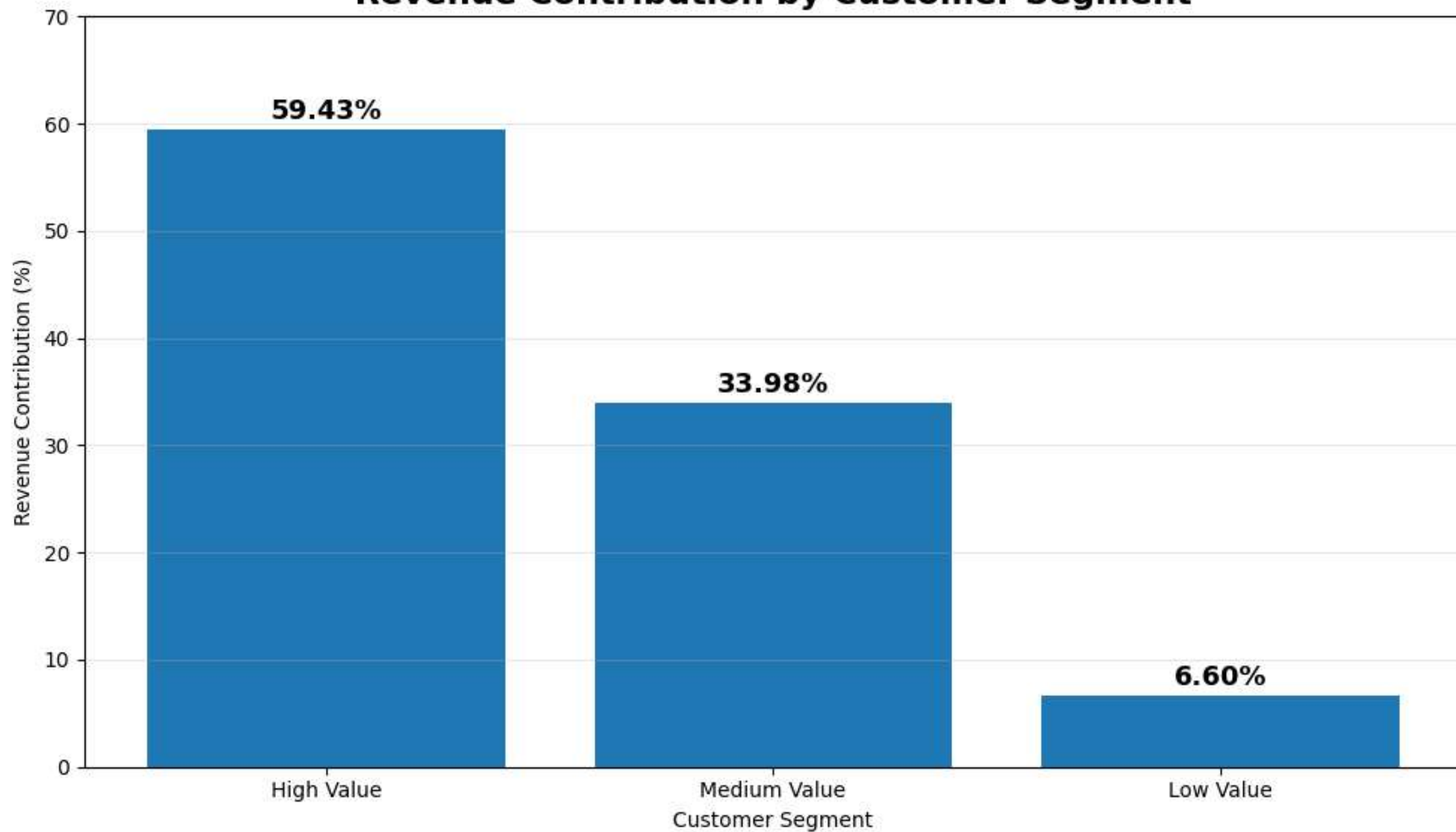




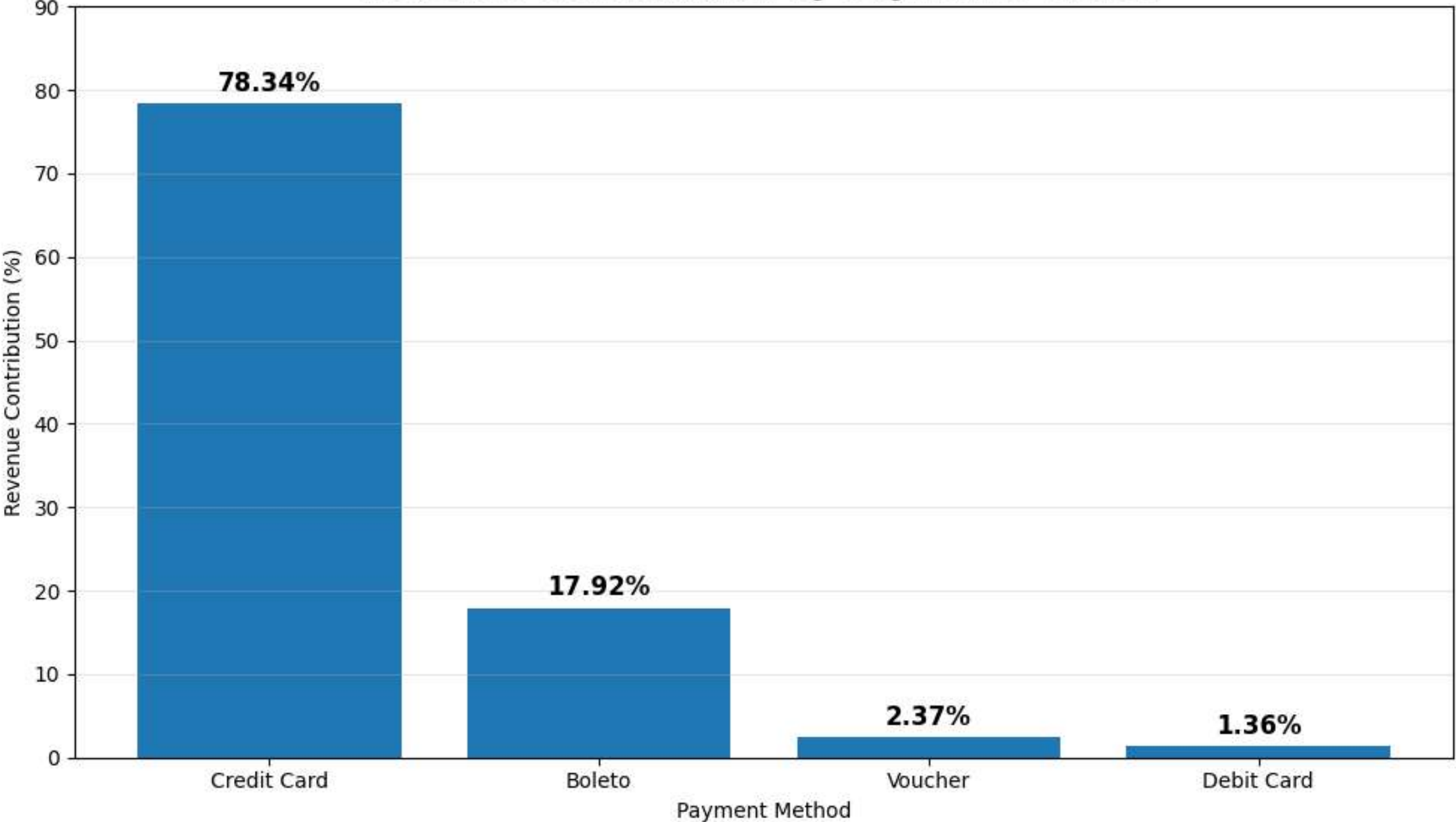
Monthly Order Trend



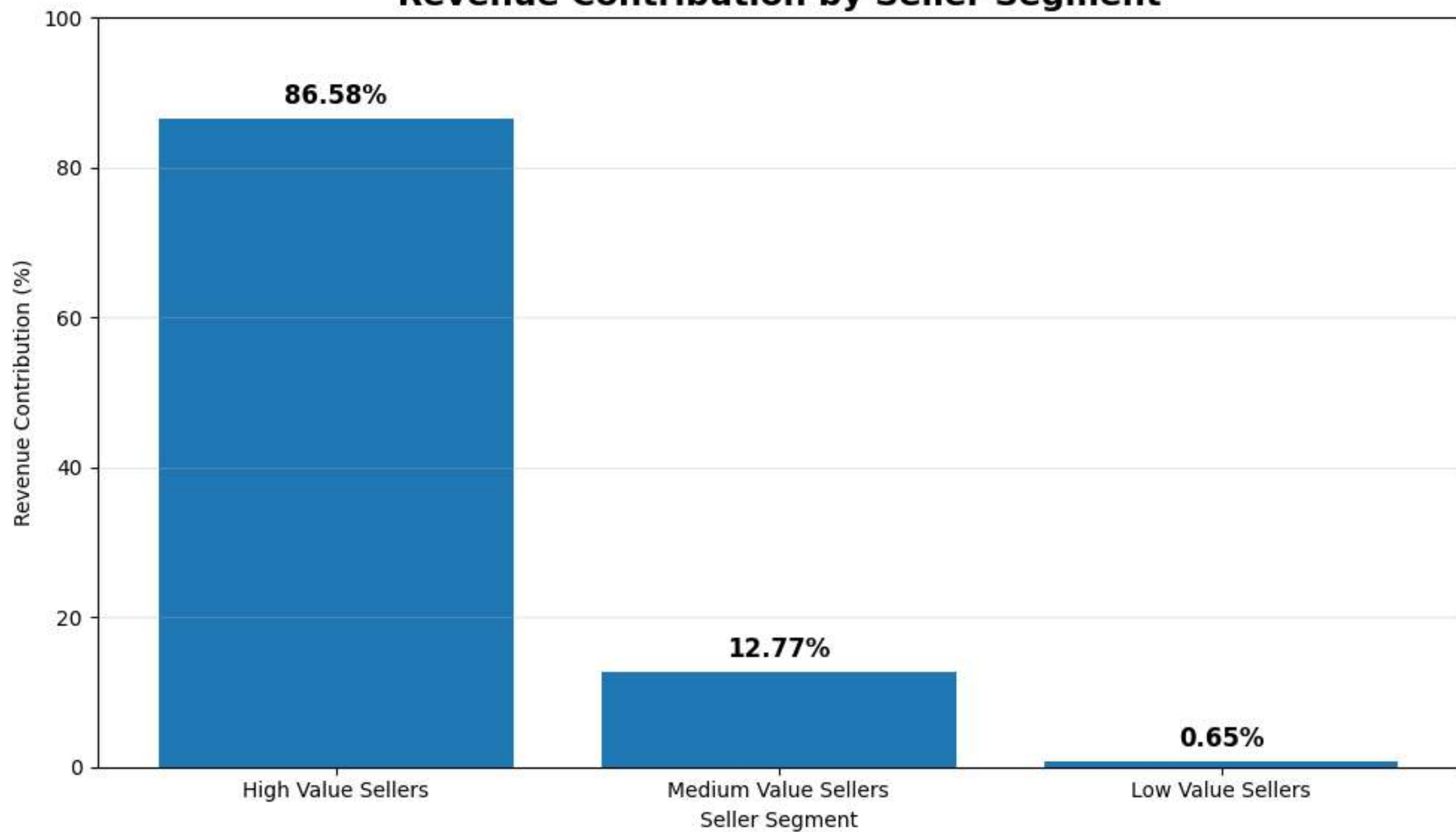
Revenue Contribution by Customer Segment



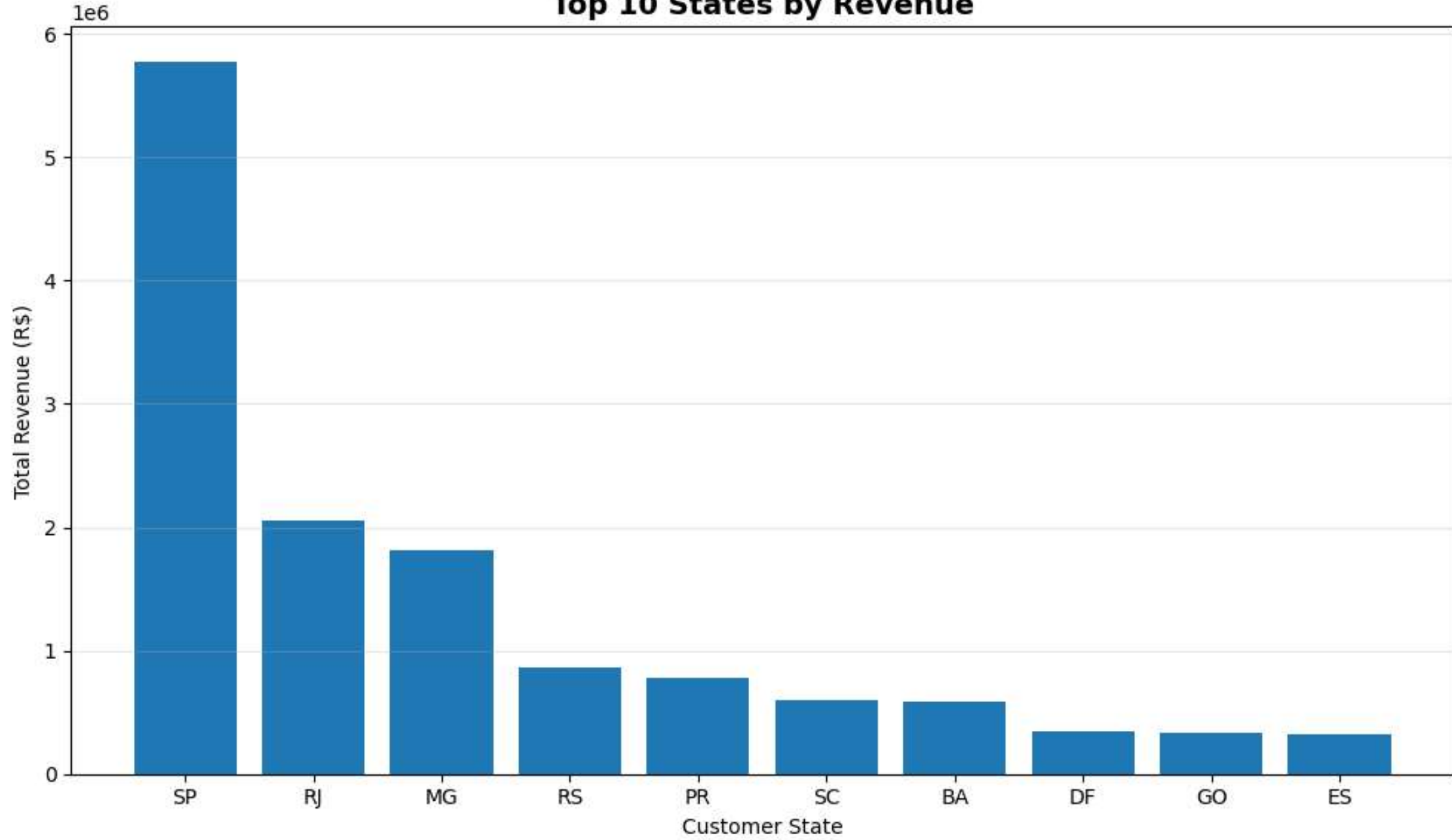
Revenue Contribution by Payment Method



Revenue Contribution by Seller Segment



Top 10 States by Revenue



Top 10 Product Categories by Revenue

